

Environment Similarity Index: Quantifying map-environment divergences for robot localization performance estimation and improvement

Christian Rosendahl Dam¹, Daniel Suvei²
and Leon Bodenhagen³

Abstract—Should take about this much space:

[illegible]

I. INTRODUCTION

Robust, lifelong localization is a fundamental ability for all Autonomous Mobile Robots (AMRs). For many years, the gold standard to achieve this has been to use some variation of Monte Carlo Localization, as it strikes a good balance between cost and performance. It can perform global localization, and is multi-modal. For indoor industrial applications, it works well in many cases. It does however, have a major drawback: It assumes the environment is static. As soon as it' surroundings have changed too much, the particle filter diverges and fails, because it does not update it's representation. Making a system that is robust to change is still an ongoing challenge (ref). Some would argue that Simultaneous Localization and Mapping (SLAM) is the best way to solve this; but SLAM suffers in reliability. No system is perfect, and sooner or later, an error will find its way into the system. The resulting erroneous pose will propagate into the resulting map, reducing accuracy or cause a complete loss of localization. Since there is no ground truth to inform the system when the errors was introduced, it is impossible to recover the last valid map state. Therefore, this paper suggests a different approach; instead of updating the map, the environment is analysed to detect changes between what is seen and what is expected. This info is then used to alter the behavior of the localization system - in this case, adaptive MCL (AMCL). To do this, we need a metric that defines

how similar (or different) the observed environment is from its representation (the map). However; to the best of our knowledge, no such metric exists. We hypothesize that such a metric can be used to infer the performance of any given localization module that takes the static world assumption, and that it can be used online to adjust its behaviour, thereby reducing the pose error. Furthermore, it can be used to generate a heatmap of an environment, which can then be interpreted by a human operator, in order to identify local regions of high divergence. These are the contributions of this paper:

We propose the Environment Similarity Index (ESI): An index that quantifies how well the currently observed environment matches the map. We also show how the ESI correlates to changes in the environment, both on simulated and real-world data, and how these changes impacts AMCL’s performance. In addition, we show how the ESI can be used by a user to easily identify areas in their site that have changed since mapping, and therefore may increase localization errors. Finally, we show how the ESI can be used to improve AMCL’s performance by using it to dynamically tune AMCL’s parameters.

II. RELATED WORK

- Bayes filtering / Probabilistic map update - Scan-BIM
- divergences - Occupancy grid vs feature maps -

III. METHODOLOGY

This section describes how changes in the environment are defined, and how they are used to calculate the ESI.

A. Changes

The difference that is measured is defined as a *change in how the robot perceives its environment*. The change is measured against the map the robot has of said environment. This is an important definition, with three major aspects. The first is that there can be a lot of changes to an environment, but if the robot does not sense it, then the changes by definition have no impact on the robot's ability to localize itself. The second is that it makes the index agnostic to which localization method is used, and which sensor modalities are available. A difference in observed visual features (I.e. SIFT) can be used just as well as geometric lidar, radar or sonar points, both 2D and 3D. The third is that changes that *could* be sensed but aren't, should not affect the

*This work has been funded by the Danish Innovation Fund.

¹Christian Rosendahl Dam is with University of Southern Denmark, The Maersk Mc-Kinney Møller Institute, Odense, Denmark
chrdam@mmmi.sdu.dk

²Daniel Suvei is with Teradyne Robotics, Odense, Denmark
dsu@mir-robots.com

³Leon Bodenhagen is with University of Southern Denmark, The Maersk Mc-Kinney Moller Institute, Odense, Denmark lebo@mmmi.sdu.dk

index. Changes that happen outside the sensor coverage can naturally not be detected, and so they have no effect on localization. Unseen changes may also change back to the initial state before they are even noticed. The metric should as mentioned apply to any kind of sensors used, but since MiR robots localize via 2D Lidar, this will be the sensor modality used from here on. The reference map could likewise be of any kind (graph based, geometry based, grid based, etc), but for this project, the (occupancy) grid-based map is selected.

Any physical object can result in a change in the environment by being either removed, added, moved, deformed or alter how it interacts with the sensor. However, this can be simplified to just Removed and Added. Any of the last three changes (moved, deformed, different sensor interaction) can be considered as a combination of removing the original item, and adding in the new. The distinction makes no difference from the robots point of view. Finally, an object can be observed to be exactly as expected - this will be denoted as Matched. There are many ways a sensor reading can be categorized as either of these, and the method chosen is by no means the only viable option.

B. The Environment Similarity Index (ESI)

The index is a ratio, based on the current pose estimate from the robots localization module - for this paper, AMCL. On every laser-scan callback, each ray point is evaluated: Is it close to an expected obstacle in the map? If yes, count as a Matched; else, count as an Added. This tells us that an object has been added to the environment that is not on the map. In addition, all laser rays that returned a value are traced in the map from the scanner and outwards to detect obstacles that they *should* have hit but didn't. These objects count towards Removed to indicate that something on the map has been removed from the environment. These classifications are then used to calculate the ESI score; Figure X provides an overview of the classification.

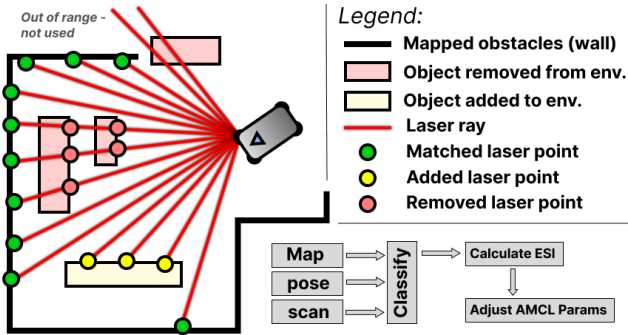


Fig. 1. Example of a figure caption.

C. Formal Definition

Matched: Let S be a 1D vector representing a single scan containing $n_s = |S|$ geometric scan points $s : S = (s_0, s_1, \dots, s_{n_s-1})$. Only scan points that return a value within min and max range are considered valid. The points

are indexed by i where $0 \leq i < n_s$. Let G be a 1D vector representing the discrete geometric reference map, flattened from its original N -dimensional form:

$G = (g_0, g_1, \dots, g_{n_g-1})$ The occupancy value at index i is $v(g_i)$, which equals 1 if g_i represents an obstacle, and 0 otherwise. The assignment $\lambda(i)$ returns the index in G with the shortest distance to the scan point s_i :

$$\lambda(i) = \arg \min_{0 \leq j < n_g} d(s_i, g_j) \quad (1)$$

The distance $d(p_1, p_2)$ is the Euclidean distance between two points in R^N , where N depends on the map representation, but any distance metric can be used. Let T be a variable distance threshold. It is based on the expected yaw noise γ [rad] and the range of the scan point $r(s_i)$:

$$T(i) = \sin(\gamma) * r(s_i) \quad (2)$$

Define a match as:

$$\chi_M(i) = \begin{cases} 1 & \text{if } d(s_i, g_{\lambda(i)}) < T(i), \\ 0 & \text{otherwise.} \end{cases} \quad (3)$$

The number of matches for a single scan is then:

$$M = \sum_{i=0}^{n_s-1} \chi_M(i) \quad (4)$$

Added: All valid scan points are classified as either Match or Added. Therefore, the number of added is simply the number of valid scan points minus the number of matches:

$$A = n_s - M \quad (5)$$

Removed: Each valid scan point defines the endpoint of a line from the scanner to that point, as seen in the world frame. The laser ray L contains all the discrete map points in G that lie on that line: $L = (l_0, l_1, \dots, l_{n_r-1})$ where l_0 is closest to the scanner. We define a removed point to be every rising edge (going from free to occupied) that the ray encounters from the scanner and out. Define removed as:

$$\chi_R(j) = \begin{cases} 1 & \text{if } v(l_j) = 1 \text{ and } v(l_{j-1}) = 0 \\ 0 & \text{otherwise.} \end{cases} \quad (6)$$

The count of removed items (first encounters) in a scan with k rays is then:

$$R = \sum_k \sum_{j=1}^{n_r^{(k)}-1} \chi_R^{(k)}(j) \quad (7)$$

ESI: The ESI score for a single scan S is then defined as:

$$ESI \in [0, 1] = \frac{M}{M + A + R} \quad (8)$$

and since $n_s = M + A$, the final expression becomes:

$$ESI = \frac{\sum_{i=0}^{n_s-1} \chi_M(i)}{n_s + \sum_k \sum_{j=1}^{n_r^{(k)}-1} \chi_R^{(k)}(j)} \quad (9)$$

If the scan only sees matches, the ratio is $1/1 = 1$, indicating a perfect match with the map. Both added and

removed points increases the denominator equally, driving the index down. Note how this reflects reality; there is no upper limit to how much you can change an environment; you can always use a sensor with a greater range and higher resolution and then add more items that are not mapped. The index therefore converges towards zero as more changes are made, but it will only be exactly zero if it sees zero matches. The index can be calculated for any number of measurements per sensor update, and can be used for any given localization module.

IV. EXPERIMENTS

In this section, we describe the two experiments designed demonstrate how the ESI can be used to:

- Quantify and visualize changes in the environment, and to what degree it accurately reflects these changes.
- Estimate and predict AMCL's performance in rapidly changing environments.
- Reduce AMCL's pose estimation error in rapidly changing environments by dynamically tuning its parameters.

A. Real-world experiment

To demonstrate that the ESI works on a real robot in a real environment, and that it can be used to visualize changes in it, we used a MiR100 AMR as the platform. It has two SICK S300 safety rated 2D Lidar sensors that scans in a horizontal plane 20cm above the ground, with a 360 degree field of view and a max range of 30m. The venue is the production hall of a packaging and labeling company. It was chosen because it represents the type of environment where AMCL often struggles; it is constantly developing, and most of the time, the the majority of what the robot is able to see consists of transient objects, as seen in Figure X.

1) **Setup:** A small section of the hall was mapped, using the native mapping functionality (based on ROS1 cartographer). Two waypoints, Start and Finish, were placed. The task of the robot was simply to move from Start to Finish. In order avoid disrupting their production, instead of moving the items around, we first recorded the map, and then gradually edited it manually, resulting in similar map-environment disagreements. Some transient objects were removed, and new ones added. The robot was tested in four scenarios: 1) No, 2) light, 3) moderate and 4) severe changes. Each scenario was executed 5 times, and the ESI recorded for all runs. When the robot reached the end, its pose error was obtained from a reference coordinate system on the floor. The ESI scores found during each run were then used to create heatmaps for each scenario, as seen in figure X. The method used is Inverse Distance Weighting (IDW) with a power of 2 and radius of 30m. A Line-of-sight (LoS) check was implemented in order to reflect how readings on either side of an obstacle should not affect each other.

TABLE I

ESI RESULTS FOR THE REAL-WORLD EXPERIMENT. 1) NO CHANGES, 2) LIGHT CHANGES, 3) MODERATE CHANGES, 4) SIGNIFICANT CHANGES.

ROWS MARKED WITH * HAVE UNITS [mm] AND [°], RESPECTIVELY.

Scenario	1)		2)		3)		4)	
* Pose RMSE	22.4	2.4	17.5	2.1	21.1	2.4	N/A	N/A
* Pose Std. dev.	25.1	2.7	19.5	1.1	20.5	1.0	N/A	N/A
ESI mean	0.94		0.64		0.46		0.22	
ESI std. dev.	0.04		0.12		0.10		0.09	



Fig. 2. Photo of a MiR100 moving through the production hall towards the finish waypoint.

2) **Results:** A Levene's test of equality of variances on the position errors (Euclidean distance from goal) for scenarios 1-3 shows no significant differences in variance ($p = 0.8153$, center=median). A one-way ANOVA test shows no significant differences in mean position error ($p = 0.7046$). These tests show that for no, light and moderate changes, AMCL is able to consistently maintain a low pose error. However, when the changes became severe, AMCL failed in every run, and the robot was unable to reach the goal. This corresponded with a low mean ESI score of 0.22 for scenario 4.

The heatmap clearly shows how the ESI reflects the intensity of local changes across the hall in the different scenarios.

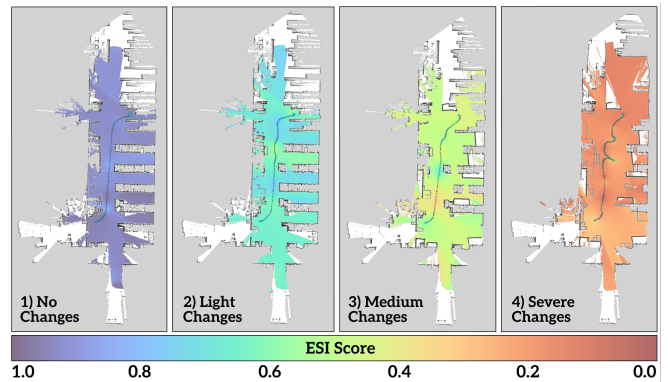


Fig. 3. ESI-based Heatmaps for each of the four scenarios in the same section of the hall.

B. Simulation experiment

...

V. CONCLUSIONS

...

VI. LIMITATIONS AND FUTURE WORK

...

APPENDIX

- Raw position data for the real-world experiment -

ACKNOWLEDGMENT

REFERENCES

- [1] M. Young, *The Technical Writer's Handbook*. Mill Valley, CA: University Science, 1989.
- [2] J. Doe, A. Smith, and B. Johnson, "Lorem ipsum dolor sit amet, consectetur adipiscing elit," in *Proc. IEEE Int. Conf. Robotics Automat.*, 2020, pp. 123–128.
- [3] C. Williams, D. Brown, and E. Davis, "Sed do eiusmod tempor incididunt ut labore," *IEEE Trans. Robot.*, vol. 35, no. 4, pp. 456–467, Aug. 2019.
- [4] F. Miller, G. Wilson, and H. Taylor, "Ut enim ad minim veniam, quis nostrud exercitation," in *Proc. IEEE/RSJ Int. Conf. Intell. Robots Syst.*, 2021, pp. 234–241.
- [5] I. Anderson, J. Martinez, and K. Lee, "Ullamco laboris nisi ut aliquip ex ea commodo consequat," *IEEE Robot. Automat. Lett.*, vol. 6, no. 2, pp. 789–795, Apr. 2021.
- [6] L. Garcia, M. Rodriguez, and N. Patel, "Duis aute irure dolor in reprehenderit," in *Proc. IEEE Int. Conf. Robot. Automat.*, 2018, pp. 567–574.
- [7] O. Thompson, P. White, and Q. Harris, "In voluptate velit esse cillum dolore eu fugiat," *IEEE Trans. Automat. Sci. Eng.*, vol. 17, no. 3, pp. 1234–1245, Jul. 2020.
- [8] R. Clark, S. Lewis, and T. Walker, "Nulla pariatur excepteur sint occaecat cupidatat," in *Proc. IEEE Conf. Decision Control*, 2019, pp. 345–352.
- [9] U. Young, V. Hall, and W. Allen, "Non proident, sunt in culpa qui officia deserunt," *IEEE Trans. Syst., Man, Cybern., Syst.*, vol. 50, no. 8, pp. 2345–2356, Aug. 2020.
- [10] X. King, Y. Wright, and Z. Scott, "Mollit anim id est laborum," in *Proc. IEEE Int. Conf. Robotics Automat.*, 2022, pp. 678–685.
- [11] A. Adams, B. Baker, and C. Carter, "Sed ut perspiciatis unde omnis iste natus error," *IEEE Robot. Automat. Mag.*, vol. 27, no. 2, pp. 89–97, Jun. 2020.
- [12] D. Evans, E. Foster, and F. Green, "Sit voluptatem accusantium doloremque laudantium," in *Proc. IEEE/RSJ Int. Conf. Intell. Robots Syst.*, 2020, pp. 456–463.
- [13] G. Hill, H. Jackson, and I. Jones, "Totam rem aperiam, eaque ipsa quae ab illo inventore," *IEEE Trans. Robot.*, vol. 36, no. 5, pp. 1234–1245, Oct. 2020.
- [14] J. Kelly, K. Lopez, and L. Moore, "Veritatis et quasi architecto beatae vitae dicta sunt," in *Proc. IEEE Int. Conf. Robot. Automat.*, 2021, pp. 789–796.
- [15] M. Nelson, N. Owens, and O. Parker, "Explicabo nemo enim ipsam voluptatem quia voluptas," *IEEE Robot. Automat. Lett.*, vol. 5, no. 4, pp. 2345–2352, Oct. 2020.
- [16] P. Quinn, R. Reed, and S. Ross, "Sit aspernatur aut odit aut fugit, sed quia consequuntur," in *Proc. IEEE Conf. Decision Control*, 2020, pp. 567–574.
- [17] T. Stewart, U. Turner, and V. Underwood, "Magni dolores eos qui ratione voluptatem sequi nesciunt," *IEEE Trans. Automat. Sci. Eng.*, vol. 18, no. 2, pp. 3456–3467, Apr. 2021.
- [18] W. Vance, X. Vega, and Y. Vincent, "Neque porro quisquam est, qui dolorem ipsum quia dolor sit amet," in *Proc. IEEE Int. Conf. Robotics Automat.*, 2019, pp. 890–897.
- [19] Z. Ward, A. Watson, and B. Webb, "Consectetur, adipisci velit, sed quia non numquam eius modi tempora," *IEEE Trans. Syst., Man, Cybern., Syst.*, vol. 51, no. 6, pp. 4567–4578, Jun. 2021.
- [20] C. West, D. Wood, and E. Wright, "Incididunt ut labore et dolore magnam aliquam quaerat voluptatem," in *Proc. IEEE/RSJ Int. Conf. Intell. Robots Syst.*, 2021, pp. 123–130.
- [21] F. Yang, G. Young, and H. Zhang, "Ut enim ad minima veniam, quis nostrum exercitationem ullam corporis," *IEEE Robot. Automat. Mag.*, vol. 28, no. 3, pp. 123–134, Sep. 2021.
- [22] I. Zhao, J. Zhou, and K. Zhu, "Suscipit laboriosam, nisi ut aliquid ex ea commodi consequatur," in *Proc. IEEE Int. Conf. Robot. Automat.*, 2020, pp. 234–241.
- [23] L. Abbott, M. Bell, and N. Cooper, "Quis autem vel eum iure reprehenderit qui in ea voluptate velit esse," *IEEE Trans. Robot.*, vol. 37, no. 1, pp. 5678–5689, Feb. 2021.
- [24] O. Davis, P. Edwards, and Q. Ellis, "Quam nihil molestiae consequatur, vel illum qui dolorem eum fugiat," in *Proc. IEEE Conf. Decision Control*, 2021, pp. 345–352.
- [25] R. Fisher, S. Ford, and T. Fox, "Quo voluptas nulla pariatur at vero eos et accusamus," *IEEE Robot. Automat. Lett.*, vol. 6, no. 3, pp. 6789–6796, Jul. 2021.

- [26] U. Grant, V. Gray, and W. Green, "Et iusto odio dignissimos ducimus qui blanditiis praesentium," in Proc. IEEE Int. Conf. Robotics Automat., 2021, pp. 456–463.
- [27] X. Hayes, Y. Hughes, and Z. Hunt, "Voluptatum deleniti atque corrupti quos dolores et quas molestias," IEEE Trans. Automat. Sci. Eng., vol. 19, no. 1, pp. 7890–7901, Jan. 2022.
- [28] A. Ingram, B. Irwin, and C. Irwin, "Excepturi sint occaecati cupiditate non provident, similique sunt in culpa," in Proc. IEEE/RSJ Int. Conf. Intell. Robots Syst., 2022, pp. 567–574.
- [29] D. Jenkins, E. Jensen, and F. Johnson, "Qui officia deserunt mollitia animi, id est laborum et dolorum fuga," IEEE Trans. Syst., Man, Cybern., Syst., vol. 52, no. 4, pp. 2345–2356, Apr. 2022.
- [30] G. Klein, H. Knight, and I. Knox, "Et harum quidem rerum facilis est et expedita distinctio," in Proc. IEEE Int. Conf. Robot. Automat., 2022, pp. 678–685.